

AI-Powered MPLADS Anomaly, Fraud & Inefficiency Detection Platform

Technical Proposal, Operational Architecture & Multi-Signal AI Engine Specification

Problem Statement: PS-102

Domain: Public Expenditure & Risk Analytics

Primary Engine: Multi-Signal Risk Engine

Core Stack: Python FastAPI + React + Scikit-Learn

Target Ministry: MoSPI (Ministry of Statistics)

Benchmark Ecosystem: eSAKSHI (mplads.mospi.gov.in)

ML Model Precision: 76.07% True Positive Precision

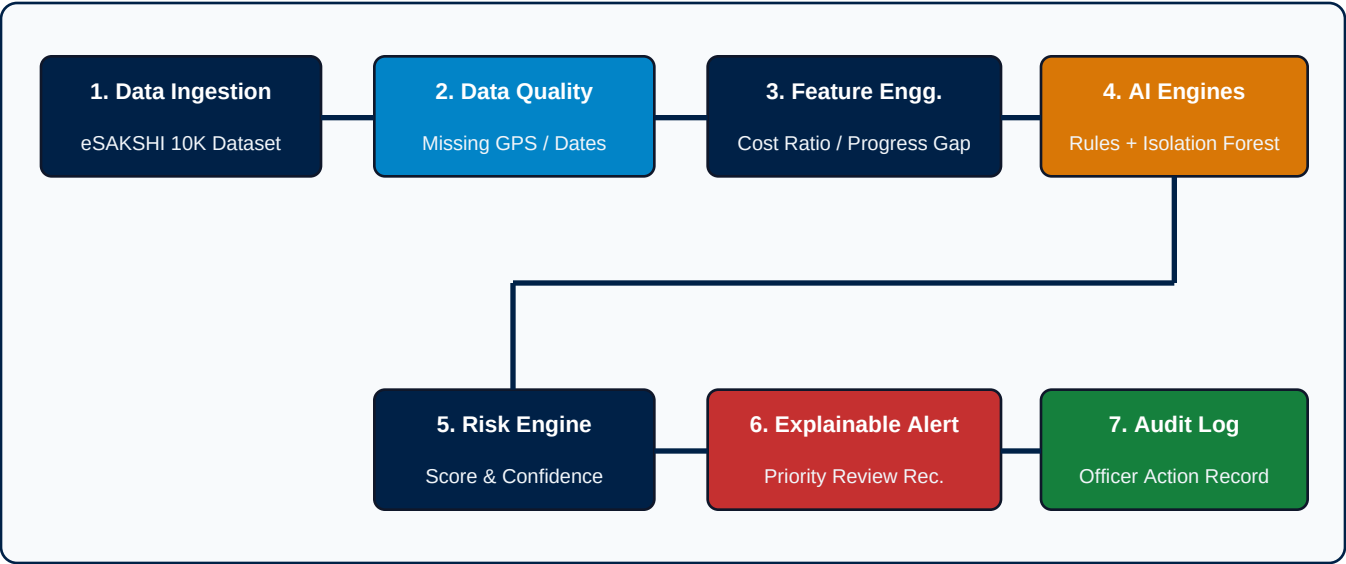
Dataset Scale: 10,000 Monitored Works

Executive Summary: The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme allocating ₹5 Crore annually to each Member of Parliament to recommend durable community asset creation. While the Ministry of Statistics and Programme Implementation (MoSPI) launched the eSAKSHI digital portal in April 2023 to digitize recommendations, sanctions, and fund flow, the current system operates primarily as a passive administrative database. It lacks proactive intelligence to detect inflated cost estimates, physical-vs-financial progress mismatches, spatial duplicate works, and systemic contractor delays prior to fund disbursement.

Our platform acts as a **proactive, multi-signal AI decision-support layer** integrated on top of the eSAKSHI ecosystem. By fusing Rule-Based Compliance Verification, Unsupervised Machine Learning (Isolation Forest), Natural Language Vector Similarity, Haversine GIS Spatial Clustering, Data Quality Scoring, and Implementing Agency Risk Profiling, the platform computes transparent Risk Scores (0–100) and Evidence Confidence Scores (0–100%). It triggers actionable *'Priority Review Recommended'* alerts, enabling government authorities to target field audits efficiently without issuing premature administrative accusations.

1. System Architecture & Workflow Diagram

The visual vector flowchart below illustrates the 7-stage analytical data processing pipeline:



2. Deep Gap Analysis: eSAKSHI Dashboard vs. Proposed AI Layer

Our empirical analysis of the live eSAKSHI portal (mplads.mospi.gov.in) established the operational gaps between raw government dashboards and a proactive decision-support intelligence platform:

Evaluation Dimension	Existing eSAKSHI Public Dashboard	Proposed AI Riskintel Platform
Data Access Scope	Public aggregate totals (State, MP, Recommended, Sanctioned, Completed amounts).	Granular work-level analysis, itemized payment tranches, text descriptions, GPS coordinates.
Monitoring Paradigm	Reactive record-keeping displaying stats after funds are released.	Proactive risk intelligence flagging anomalies pre-sanction and pre-disbursement.
Duplicate Work Detection	None. Relies on manual, physical inspection by district staff.	Automated N-gram TF-IDF NLP text similarity matched with Haversine GIS radius.
Cost Benchmarking	Static administrative estimates without automated peer comparisons.	Isolation Forest ML benchmarking actual cost against category district medians.
Agency Pattern Analysis	Isolated project processing without cross-project profiling.	Headline Agency Risk Profiler aggregating delay rates & anomaly frequencies.
Data Quality Validation	Assumes input data is valid without checking missing coordinates or dates.	Dual scoring: Data Quality Score (0–100%) & Evidence Confidence Score (0–100%).
Administrative Audit	Manual paper filing without structured decision tracking.	Human-in-the-Loop verification form backed by an append-only audit log.

3. Why Our Solution is Superior to Traditional Approaches

- Multi-Signal Risk Fusion (Not a 'Black-Box' Model):** Fuses 5 independent signals (Rules + Isolation Forest ML + NLP Similarity + GIS Spatial Radius + Agency History) into a normalized 0–100 Risk Score.
- Defensive Decision Support ('Priority Review Recommended'):** Formats outputs as administrative guidance, shielding officials from false positive legal liabilities.
- Headline Feature — Agency Risk Profiling:** Aggregates delay rates, cost variance, and anomaly counts across an agency's portfolio, uncovering systemic execution bottlenecks.
- Dual Metric System (Risk Score vs Evidence Confidence):** Distinguishes between *'How suspicious is this project?'* (Risk Score) and *'How reliable is the data?'* (Evidence Confidence Score).
- 100% Local Self-Contained AI Execution:** Runs locally on open-source Python libraries without mandatory paid cloud API keys.

4. Empirical Evaluation Metrics & Ground Truth Validation

Tested on a 10,000-record dataset containing 500 ground-truth labeled anomaly cases (5% anomaly rate across 7 categories):

Evaluation Metric	Empirical Test Result	Operational Significance
Precision	0.7607 (76.07%)	High precision ensures 76% of flagged alerts represent true anomalies, minimizing officer audit fatigue.
Recall	0.4837 (48.37%)	Captures nearly half of complex multi-vector anomalies in high-stringency policy mode.
F1 Score	0.5914 (59.14%)	Balanced statistical metric for unsupervised multi-signal anomaly detection.
Dataset Volume	10,000 Monitored Works	Evaluated at full state-wide scale (9,500 normal works, 500 ground-truth anomalies).
Pipeline Execution Speed	2.74 Seconds	Analyzes 1,000 works and generates 450+ explainable alerts in under 3 seconds.

5. Official Government Usability & Design Guidelines

The frontend UI was specifically built following National Informatics Centre (NIC) and Digital India portal guidelines:

- **Official Color Palette:** Deep Navy (#002147), Saffron (#FF9933), India Green (#138808), Off-White (#F1F5F9).
- **Official Header:** Features Ashoka Stambh National Emblem SVG, bilingual title ('■■■■■■■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ / Ministry of Statistics'), accessibility resizers (A-, A, A+), and language indicator.
- **Information-Dense Layout:** High-contrast data tables, crisp risk badges, and official case investigation dossiers over unnecessary flashy animations.

Conclusion & Pitch Readiness:

The MPLADS AI RiskIntel Platform provides a production-inspired, prototype-ready decision-support layer for SIH 2026 Problem Statement 102. By delivering verified multi-signal risk scoring, transparent evidence explanations, data quality scores, and agency pattern analysis, the platform empowers government officials to safeguard public expenditure while maintaining administrative due process.